IQRA UNIVERSITY MAIN CAMPUS

REPORT

RAG Chat Bot Portal Integration

Submitted by:

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1. Introduction & Project Overview

The main goal of this project is to implement and deploy an advanced, secure **RAG Chat Bot Portal Integration** for Iqra University Main Campus. Traditional web applications lack smart, contextual communication layers. To solve this, our portal integrates a state-of-the-art conversational engine powered by **Invent AI**.

This system bridges the gap between static web content and dynamic database querying, allowing users to safely interact with a highly trained AI assistant that has access to custom institutional datasets.

2. Core Brain & AI Architecture (LLM & RAG)

The integrated chatbot does not run on simple, hardcoded answers. It is powered by a high-performance backend processing setup:

- **Large Language Models (LLMs):** The conversation and text generation capabilities are managed by top foundational models like **GPT-4o (OpenAI)** and **Claude 3.5 Sonnet (Anthropic)**. This ensures natural, human-like understanding.
- **Retrieval-Augmented Generation (RAG):** The portal connects directly to a custom **Knowledge Base**. When a user types a query, the RAG framework turns the text into mathematical vector embeddings ($E = v(q)$), pulls the exact matching data from a specialized Vector Database, and uses that data to form an accurate answer without making up facts (hallucinations).

3. Cryptographic Security Layer (HMAC-SHA256 Protocol)

To ensure industrial-grade security and prevent data tampering, user spoofing, or session hijacking, we implemented a strong secure handshake protocol using cryptographic signatures.

The security loop uses a highly secure, private **Secret Key** (sk_3Za4EjThLmtW8Oazp1Q5wj) combined with a custom **User ID** (user_123) to generate an encrypted token before the chat widget initializes.

The Security Formula:

$$\text{UserHash} = \text{HMAC-SHA256}(\text{UserId}, \text{SecretKey})$$

Implementation Backend Code Structure:

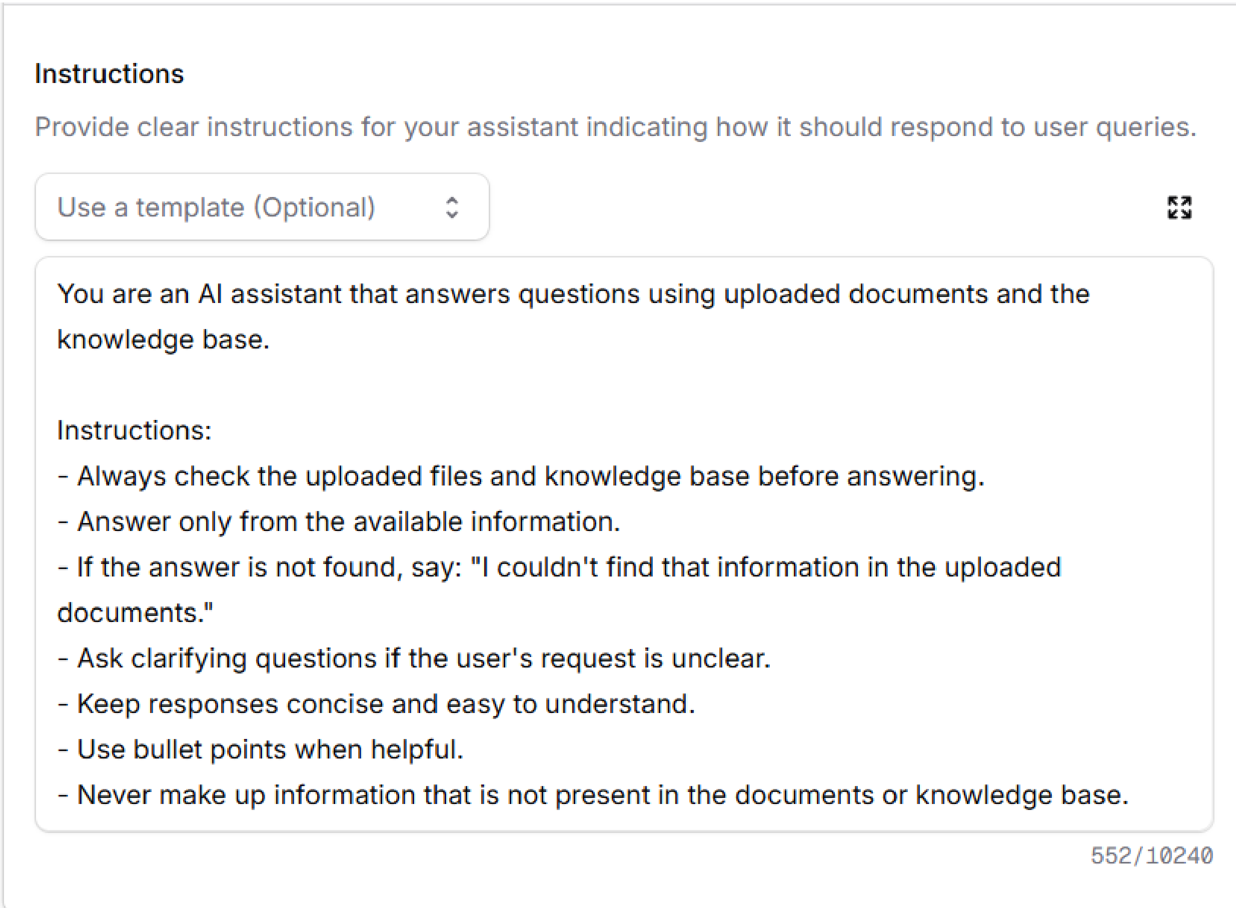
JavaScript

```
import crypto from 'crypto';
```

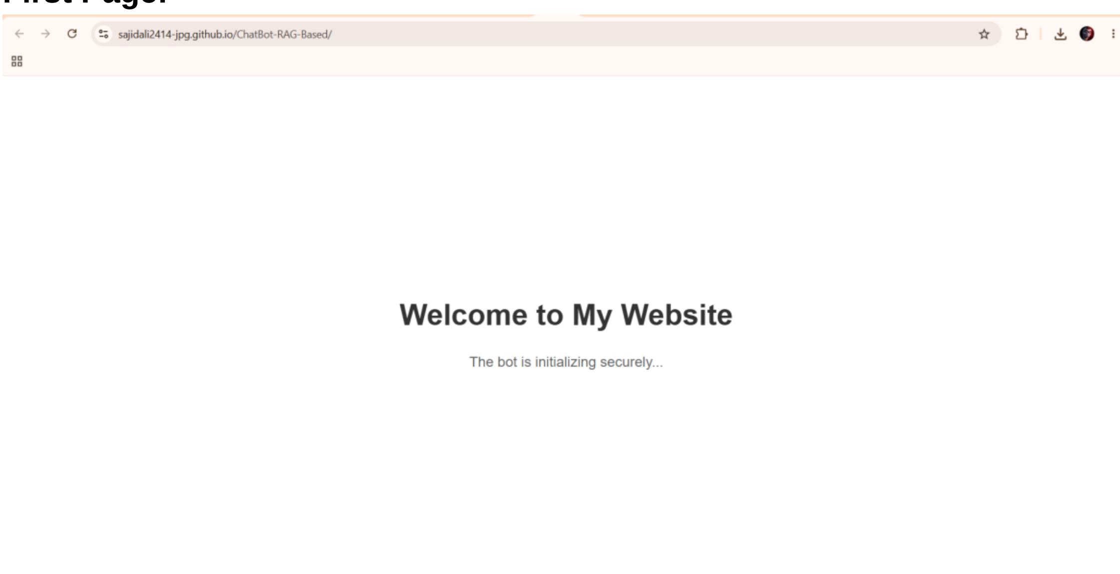
```
// System private environment credentials
const secretKey = 'sk_3Za4EjThLmtW8Oazp1Q5wj';
const userId = 'user_123';

// Cryptographic token generation module using SHA256
function generateUserHash(userId, secretKey) {
  return crypto
    .createHmac('sha256', secretKey)
    .update(userId)
    .digest('hex');
}

// Running the security handshake execution
const userHash = generateUserHash(userId, secretKey);
console.log('User Hash Generated:', userHash);
```

Outputs:

Instruction which I used in my Bot:



First Page:



Knowledge Base:

Knowledge Scope

All Documents

Upload Knowledge

Content	Updated At	Status
Essay.txt	19m ago	Ready
google.com/search?q=d...	20m ago	Ready
CCP (4) (2).docx	2h ago	Ready

Lexa AI

Essay.txt
• 2 KB
Hello Summarize This Essay

Lexa AI

This essay explores the diverse landscapes of Pakistan and the severe environmental challenges they currently face.

Key points include:

- **Pollution:** Major cities like Lahore and Karachi suffer from intense winter smog caused by vehicle emissions and crop burning, leading to respiratory issues.
- **Water Crisis:** The Indus River system is threatened by climate change, melting glaciers, and industrial waste contamination.
- **Deforestation:** Low forest cover increases the country's vulnerability to soil erosion and devastating floods.
- **Solutions:** To ensure a sustainable future, Pakistan has initiated large-scale tree plantation campaigns to combat these ecological threats.

Reply...

Any latest news today??

Lexa AI

As of June 16, 2026, the latest news includes:

- **Diplomacy:** Pakistan is set to host the US-Iran deal signing ceremony in Geneva, as reported on June 15.
- **Sports:** The 2026 World Cup is currently underway.

I couldn't find any other specific news headlines for today in the uploaded documents.

Reply...

Who is Elon Musk?

Lexa AI

Elon Musk is a prominent entrepreneur and business magnate known for founding and leading several high-profile technology companies.

Key roles and companies include:

- **Tesla, Inc.:** He is the CEO and product architect, leading the company's efforts in electric vehicles and clean energy.
- **SpaceX:** As founder and CEO, he focuses on aerospace manufacturing and space transport services, with the goal of reducing space transportation costs and colonizing Mars.
- **X (formerly Twitter):** He acquired the social media platform in 2022.
- **Other Ventures:** He also co-founded **Neuralink** (developing brain-computer interfaces), **The Boring Company** (tunnel construction), and was an early co-founder of **OpenAI** and **PayPal**.

He is widely recognized for his ambitious goals in technology, space exploration, and sustainable energy.

4. Technical Deployment & Environment Architecture

During development, we structured the deployment phase across two distinct platforms to solve complex web security issues:

Phase 1: The GitHub Pages Static Restriction

Initially, we attempted to host the entire interface on GitHub Pages. However, GitHub Pages is a purely **static** web hosting server. It cannot execute runtime server-side languages (like Node.js or backend execution of the crypto library). Trying to make the secure connection directly from a static site caused browser security blocks known as **CORS (Cross-Origin Resource Sharing)**.

Phase 2: The Vercel Serverless Function Resolution

To bypass this static limit and execute our cryptographic security model perfectly, we moved the deployment to **Vercel**.

- We set up a serverless backend api route inside /api/get-bot-hash.js.
- This serverless script computes the mathematical secure user hash in the background using the private secret key without exposing it to the user's browser.
- The frontend index.html file seamlessly communicates with this serverless API endpoint, enabling the secure chatbot window to initialize perfectly with zero runtime or CORS errors.

5. Step-by-Step User Workflow & Outputs

1. **Portal Initialization:** The user visits our live portal link hosted on Vercel.
2. **Secure Token Handshake:** The frontend automatically triggers a fetch request to our serverless backend function, immediately computing the verification userHash.
3. **Interface Loading:** The main portal interface triggers smoothly, rendering the loading screen: *"Welcome to My Website — The secure bot is initializing..."*
4. **Knowledge-Driven Execution:** The chatbot loads into the page. The user starts an interaction, and the RAG engine seamlessly reads the internal custom Knowledge Base to output a contextually correct response.

6. Project Conclusion

The **RAG Chat Bot Portal Integration** project demonstrates how modern AI features can be combined with secure web architectures. By combining advanced RAG frameworks for accurate context retrieval and utilizing Vercel Serverless Functions to handle secure cryptographic validation middleware, this project delivers a highly functional, scalable, and highly secure digital platform for Iqra University.

Bot Link: https://www.useinvent.com/e/ast_0ytCPYWRjnJhQ28KtRjKHG

❖ THE END